CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 5: Project Development Phase

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 5: Project Development Phase**
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Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Snowflake SQL Pipeline Implementation

The complete end-to-end data pipeline was developed and structured into modular, production-grade SQL scripts checked into the project repository under sql/:

Script File	Module Role	Implementation Details & SQL Logic
sql/01_data_architecture_setup.sql	**Warehouse & Storage Setup**	Provisions ORDERPULSE_WH (X-Small, 60s auto-suspend), ORDERPULSE_DB, schemas (RAW, STAGING, CURATED), internal stage @ORDERPULSE_STAGE, and CSV_FILE_FORMAT.
sql/02_data_ingestion_raw.sql	**Raw Landing Layer Ingestion**	Defines RAW_CUSTOMERS, RAW_RESTAURANTS, RAW_MENU_ITEMS, RAW_ORDERS, RAW_ORDER_ITEMS with permissive VARCHAR types. Executes COPY INTO commands and row validations.
sql/03_data_preparation_staging.sql	**Data Cleansing & Staging**	Cleanses data with TRY_TO_DATE, TRY_TO_TIMESTAMP_NTZ, and DECIMAL(10,2) casts. Implements TRIM(), INITCAP(), and computes is_realized and cancellation indicators.
sql/04_data_modelling_curated.sql	**Curated Star Schema & Marts**	Builds FACT_ORDERS, FACT_ORDER_ITEMS, DIM_CUSTOMERS, DIM_RESTAURANTS, DIM_MENU_ITEMS, DIM_DATE, and DIM_TIME with MD5 surrogate keys and persona data marts.
sql/05_business_analytics_kpis.sql	**Analytical KPI Queries**	Houses advanced analytical queries computing GMV, Realized Revenue, AOV, Repeat Buyer rates, Cuisine rankings, Rating elasticity, and Hourly demand curves.
sql/06_performance_testing.sql	**Performance Benchmarking**	Contains partition pruning validation queries, micro-partition inspection, query history profiling, and execution cache benchmarks.

2. Streamlit Application Architecture & Modules

The self-service presentation layer is implemented as an interactive, multi-page dashboard in streamlit_app/app.py:

- **Tab 1: Executive Overview (Grace):** Top-line GMV (■96.22L), Realized Revenue (■89.38L), Cancellation loss (■6.85L), Monthly revenue trend line, and city-wise fulfillment breakdown.
- **Tab 2: Customer Retention (Priya):** Repeat customer ratio donut chart (69.8% repeat vs 30.2% single), RFM loyalty tier segmentation, and basket size spend elasticity curve.
- **Tab 3: Restaurant Partnerships (Daniel):** Restaurant rating vs. order volume elasticity scatter plot, top 10 revenue-generating cuisines (North Indian, Biryani, Chinese), and popular dish rankings.
- **Tab 4: Delivery & Operations:** City-level on-time SLA compliance matrix (81.93% overall), late delivery hotspots during peak dinner hours (7–10 PM), and root-cause cancellation analysis.
- **Tab 5: Architecture & Testing:** Interactive data lineage viewer (RAW ➔ STAGING ➔ CURATED), schema table explorer, and live partition pruning speedup benchmark visualizer.

3. Dual-Mode Database Connectivity

The Streamlit application features a resilient dual-mode architecture: it connects directly to Snowflake using snowflake-connector-python when credentials are configured, and gracefully falls back to the embedded local SQLite analytical database (data/orderpulse_local.db) for instant local demonstration with zero external dependencies.